Project "Cable Cut"

Resilience Audit, Troubleshooting, and Network Redesign

Consulting Team: 3IT-11
Date: November 8, 2026
Client: Manila BPO & Call Center (200 Seats)

Part 1: Incident Analysis

In early October 2026, a major submarine cable cut along the Philippines–Singapore route severely disrupted nationwide internet connectivity. Major telecommunications providers, including Globe, PLDT, and Converge, were forced to rapidly reroute traffic, with Globe officially projecting that complete physical repairs would take "several weeks" [1].

A critical lesson from this incident is the illusion of redundancy. While the Philippines boasts approximately 11 cable systems, true geographical resilience is lacking. Many of these subsea cables converge at shared physical landing stations in high-risk zones such as Batangas, Cavite, and La Union [2]. A single localized disaster—such as a typhoon or maritime accident—can therefore sever multiple seemingly independent links simultaneously.

Impact on the BPO Client: For a 200-seat BPO facility reliant on foreign clients, this lack of geographic redundancy is catastrophic. Services hosted abroad degrade instantly. During the October outage, the client experienced immediate and severe latency hikes. Real-time VoIP quality degraded to the point of dropped calls, fundamentally halting the BPO's core revenue-generating operations. Furthermore, offshore Software as a Service (SaaS) CRM platforms became intermittent, preventing agents from accessing customer records. The BPO's existing architecture, which heavily depended on continuous WAN connectivity to offshore public clouds without local edge failovers, proved to be its weakest link.

Part 2: Documentation and Baseline

Comprehensive network documentation is the foundation of the three-stage troubleshooting process. Prior to the injected faults, the following as-built records and baseline plans were established.

Logical Configuration Sheet (As-Built)

Device Name	Role / OS	IP Address	Subnet Mask	Default Gateway
R1	Edge Router (Branch)	192.168.10.1 (G0/0.10)	255.255.255.0	N/A (WAN: 203.0.113.2)
SW1 / SW2	Dist / Access Switch	192.168.99.2 / .3	255.255.255.0	192.168.99.1

Device Name	Role / OS	IP Address	Subnet Mask	Default Gateway
PC-Sales1 (VLAN 10)	Windows OS	192.168.10.11	255.255.255.0	192.168.10.1
PC-Admin (VLAN 20)	Windows OS	192.168.20.11	255.255.255.0	192.168.20.1
Local-Server	DNS/Web Server	192.168.20.10	255.255.255.0	192.168.20.1

Performance Baseline Plan

The baseline determines the "personality" of the network under normal conditions [3]. We established a 7-day baseline duration, tracking metrics every hour via SNMP.

- **Metrics Captured:** CPU utilization on R1 and SW1, Link utilization on the WAN interface (G0/1), and average ICMP latency to the Local-Server and the simulated Singapore Cloud (8.8.8.8).
- **Environmental Constraint:** Baselines must exclusively measure normal traffic patterns. Measuring during a typhoon, or during the active October cable cut routing disruption, is strictly avoided as it establishes an abnormal, highly-congested network state as the baseline standard [3].

Part 3: Troubleshooting Case File

Using the three-stage methodology (Gather, Isolate, Correct), our team successfully resolved five injected OSI layer faults within the branch network.

Ticket 1: Physical Layer — "Sales1's computer has no network at all."

Questions Asked: 1. Are the port lights illuminated? 2. Was the cable recently moved? 3. What does the network icon in the system tray show?
Ownership: Inside organization.
Method: *Bottom-Up*. Given the total absence of connectivity, starting at the physical layer (Layer 1) is the most efficient way to isolate cabling faults.
Evidence: Interface Fa0/2 on SW2 showed a down/down state. Visual inspection revealed a Copper Cross-Over cable connecting the PC to the switch.
Fix & Test: Replaced the cross-over cable with a straight-through cable. Tested by pinging 192.168.10.1; ping was successful. Documented cable standard update.

Ticket 2: Data Link Layer — "Admin can't connect to anything."

Questions Asked: 1. What is your current IP address? 2. Can you ping the Local-Server? 3. Are there any error messages?
Ownership: Inside organization.
Method: *Divide-and-Conquer*. Pinged the default gateway (192.168.20.1) which failed. Moved down to Layer 2 to check switch interconnectivity.
Evidence: `show interfaces trunk` on SW1 revealed `switchport trunk allowed vlan 10,99`. VLAN 20 (Admin) was missing from the allowed list.
Fix & Test: Issued `switchport trunk allowed vlan add 20` on SW1 Fa0/2. PC-Admin successfully acquired a DHCP lease and pinged the gateway.

Ticket 3: Network Layer — "Sales2 can ping the router but can't reach other networks."

Questions Asked: 1. Can you ping 8.8.8.8? 2. Has your IP configuration changed? 3. What is your `ipconfig` output?
Ownership: Inside organization.
Method: *Divide-and-Conquer*. Since local pings to the router interface succeeded, the physical and data link layers were functional. The issue was isolated to Layer 3 routing/addressing.
Evidence: `ipconfig` showed Default Gateway as 192.168.10.2. The actual configured IP on R1 G0/0.10 is 192.168.10.1.
Fix & Test: Corrected the PC-Sales2 IPv4 Default Gateway to 192.168.10.1. Verified by executing `tracert 8.8.8.8` which successfully routed past the local gateway.

Ticket 4: Network/NAT Layer — "IT-PC can reach the server but not the foreign server (8.8.8.8)."

Questions Asked: 1. Are local VLAN pings successful? 2. At which hop does the traceroute fail? 3. Can other VLANs reach the outside?
Ownership: Inside organization.
Method: *Top-Down* (from Tracert app down to Layer 3 NAT rules).
Evidence: `show access-lists` on R1 showed `permit 192.168.10.0` and `permit 192.168.20.0`, but lacked an entry for the IT-PC network (192.168.99.0).
Fix & Test: Added `access-list 1 permit 192.168.99.0 0.0.0.255` on R1. IT-PC immediately received NAT translation and pinged 8.8.8.8 successfully.

Ticket 5: Application Layer — "Can't open portal.company.local, but the IP works."

1. Is the URL spelled correctly? 2. Have you cleared the DNS cache? 3. Can you reach it via 192.168.20.10? Inside organization. . Because the IP worked but the domain name failed, the physical/network layers were intact, isolating the fault squarely at Layer 7 (DNS). Checked DNS services on Local-Server. The A Record for `portal.company.local` was incorrectly pointing to `192.168.20.20`. Updated the DNS A Record to `192.168.20.10`. PC successfully resolved and loaded the portal webpage. **Questions Asked:**
Ownership:
Method:*Top-Down*
Evidence:
Fix & Test:

IP SLA Verification: We configured an ICMP echo IP SLA on R1 targeting 8.8.8.8. Prior to fixing Ticket 4, the SLA reported 0% success for VLAN 99 traffic. Post-fix, `show ip sla statistics` confirmed a 100% success rate with an average RTT of 24ms, establishing continuous WAN path monitoring.

Part 4: Evolution Redesign

To ensure this BPO client survives a repeat of the October 2026 cable cut, we propose the following resilient architectural redesign based on modern network evolution principles.

IoT and Fog Computing (Edge Resiliency)

The BPO relies on real-time voice data. We will map devices to the Cisco IoT System's six pillars, focusing heavily on the **Fog Computing Pillar** [4]. Currently, all VoIP SIP sessions terminate at an offshore cloud PBX. We will deploy an edge IoT gateway/local SIP survivability server on-premises. If the WAN fails entirely, the Fog application will continue to process local routing, queue voice mail, and maintain internal station-to-station BPO communications locally rather than halting the entire floor.

Cloud Computing Strategy

We propose transitioning the BPO to a **Hybrid Cloud** model [4]:

- **Private Cloud (On-Premises):** Sensitive client financial and healthcare data will remain on local, highly secured servers to satisfy data-sovereignty laws.
- **Public Cloud (SaaS/IaaS):** Standard workloads (email, workforce management, general CRM) will utilize SaaS. This transitions heavy server capital expenditures (CapEx) into flexible, pay-as-you-go operating expenses (OpEx).

Server Virtualization

To reduce power consumption and improve disaster recovery, the BPO's 15 aging physical servers will be consolidated using a **Type 1 (Bare Metal) Hypervisor** (e.g., VMware ESXi) [4]. The hypervisor sits directly on the hardware, improving performance and robustness. Server count will drop from 15 standalone boxes to just 3 physical host servers. If one host fails during a local crisis, the management console will automatically migrate the virtual machines (VMs) to the surviving hardware, ensuring zero downtime for the voice recording software.

Software-Defined Networking (SDN)

Traditional routers process decisions and forward traffic on the same device. We will implement SDN to centralize the **Control Plane** [4]. Using an APIC-EM controller:

- **Northbound APIs:** Will communicate with the BPO's business applications, allowing the network to dynamically assign the highest Quality of Service (QoS) queues to VoIP traffic during bandwidth degradation.
- **Southbound APIs:** (e.g., OpenFlow) will push forwarding instructions down to the data plane of the physical switches. During a subsea cable failure, the controller will detect the SLA drop and instantly reprogram the flow tables across all edge devices to route traffic out of the backup LTE connection.

WAN Resilience and Diversity

Relying on cables landing in Batangas is a single point of failure. We propose two distinct WAN paths: a primary PLDT enterprise fiber connection (approx. PHP 20,000/mo) landing via a northern route, and a geographically diverse backup connection using Low Earth Orbit (LEO) satellite internet (e.g., Starlink Business, approx. PHP 15,000/mo + hardware). This guarantees that a maritime cable cut will not affect the airborne satellite backup.

Part 7: Team Contribution

Team Member	Role & Contributions
Student 1 (Lead)	Incident analysis research, Logical/Physical Topology design, Presentation Slide Deck.
Student 2 (Engineer)	Packet Tracer simulation build, IP SLA configuration, Ticket 1-3 troubleshooting.
Student 3 (Architect)	Evolution Redesign (Cloud/SDN justification), Ticket 4-5 troubleshooting, Final Report compilation.

References

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